



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SABARMATHI FUELSS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

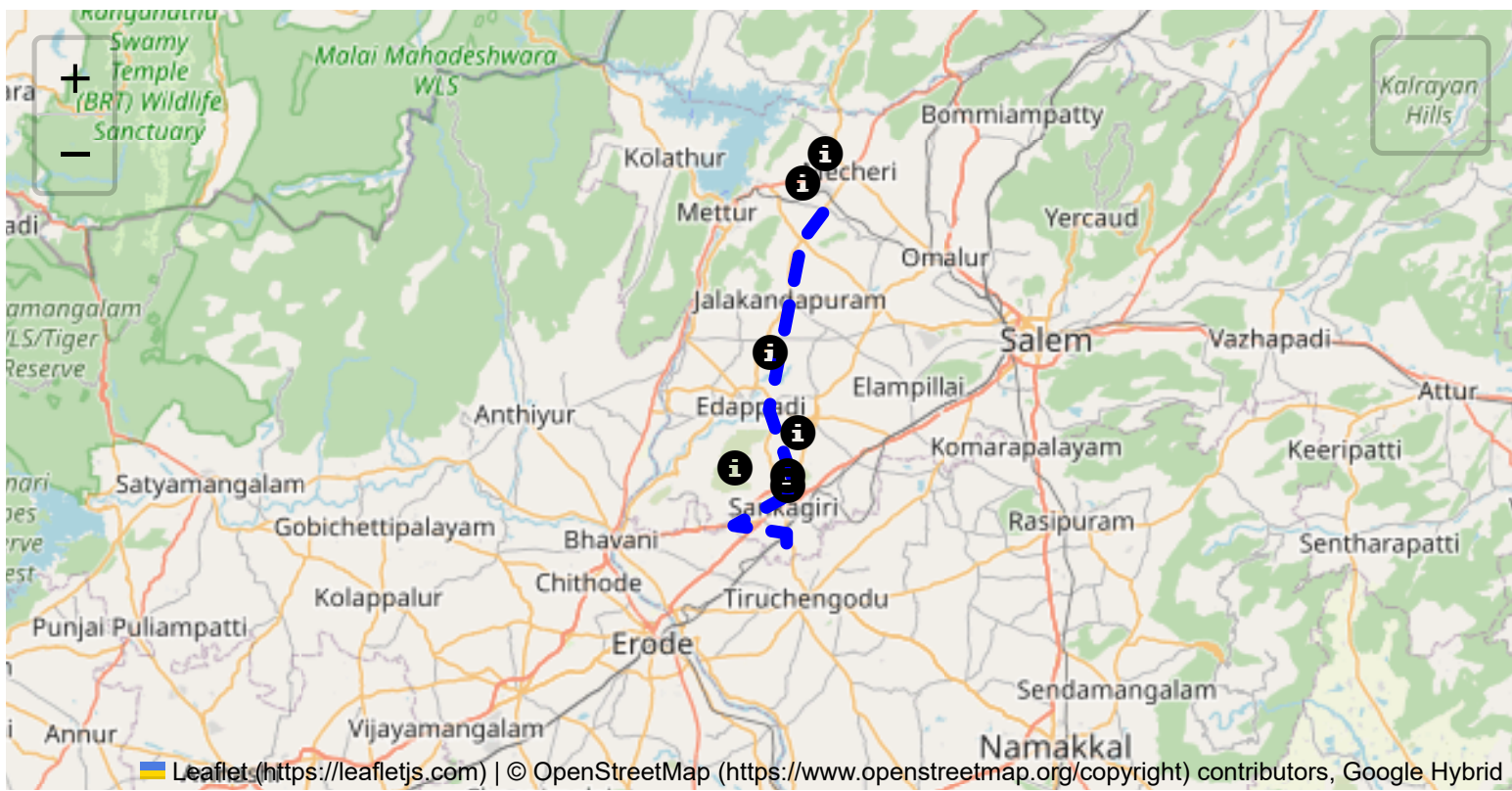
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

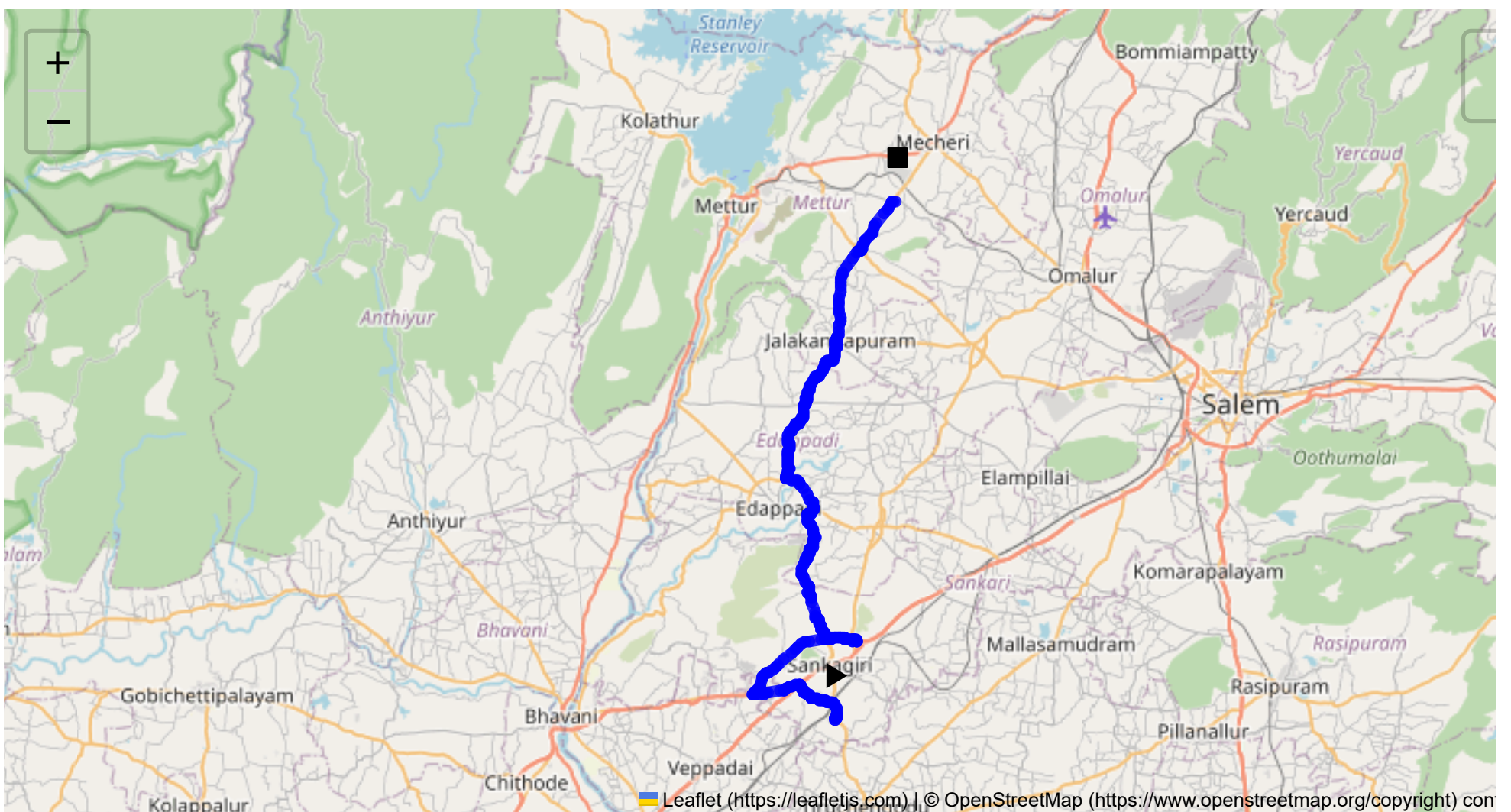
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 59.86 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.8 hours
Start: (11.4381, 77.8734)
End: (11.79325773, 77.9160782)



Welcome to the Journey Risk Management Study

Route Overview

The route covers approximately 59.86 kilometers from Sangagiri to Kuttapatti, via various points including Tamil Nadu state highway routes and local roads. The journey typically takes 1.44 hours for heavy vehicles transporting hazardous materials, passing through semi-urban and rural areas.

Weather Conditions and Hazards

The region experiences tropical weather with high temperatures in summer, and moderate levels of rainfall during the monsoon. Heavy rain can lead to waterlogging and slippery road conditions, especially from June to September. Fog can be frequent in the mornings during the winter (December-February), reducing visibility.

Traffic Patterns

Traffic congestion is common in and around the towns and junctions on the route, particularly during morning (7-9 AM) and evening (5-8 PM) rush hours. Key congestion points include:

- Sangagiri town center
- Salem-Kochi Highway intersections Due to the rural nature of parts of the road, slower-moving agricultural vehicles can create delays.

Road Quality and Infrastructure

The route comprises mixed road quality; however, major highways are generally in good condition. Smaller roads may have potholes and require cautious driving. Infrastructure like signage and street lighting is better on highways but could be lacking on rural stretches.

Alternative Routes for Emergencies

Alternative routes can be considered through NH 544 for major portions, or diverting through smaller local roads if a section is blocked. However, these may extend travel time significantly.

Regulations on Hazardous Material Transport

Local regulations require that vehicles transporting hazardous materials adhere to specific routes approved by regional transport authorities, which often include highways and designated roads. There are stringent checks especially near populated areas and entries to urban zones.

Historical Incidents

The region has seen occasional incidents involving heavy goods vehicles, mainly due to brake failures or road obstructions. No specific incidents involving hazardous materials were recorded in recent years on this route.

Environmental Considerations

The route passes through agricultural zones and near local water bodies. Ensuring no spillage or leakage during transport is crucial to prevent contamination. Noise pollution should be minimized especially near villages and town areas.

Communication Coverage

Mobile network coverage is generally good along most parts of the route, except potential dead zones in the remote rural stretches. Drivers are advised to check network availability with providers beforehand.

Emergency Response Times

In urban areas, emergency services, such as ambulances and fire services, have a response time of 15-30 minutes. In rural sections, this time can extend to 30-45 minutes.

Risk Assessment Summary

Overall, the route presents moderate risks primarily due to weather, road conditions, and potential traffic congestion. Adherence to safety protocols, regulatory compliance, and preparedness for weather changes will mitigate many risks. Planning travel outside of peak congestion times and ensuring vehicle readiness for diverse road conditions, along with awareness of emergency contacts and protocols, are recommended for minimizing potential hazards.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44898, 77.87410	30 KM/Hr
6	Turn	Medium	11.45357, 77.85789	30 KM/Hr
7	Turn	High	11.45352, 77.85698	15 KM/Hr
8	Turn	Medium	11.45847, 77.85140	30 KM/Hr
9	Turn	Medium	11.45898, 77.85135	30 KM/Hr
10	Turn	Medium	11.46303, 77.84945	30 KM/Hr
11	Turn	Medium	11.46314, 77.84928	30 KM/Hr
12	Turn	Medium	11.45543, 77.81740	30 KM/Hr
13	Turn	Medium	11.45563, 77.81708	30 KM/Hr
14	Turn	High	11.45631, 77.81668	15 KM/Hr
15	Turn	Medium	11.49377, 77.88573	30 KM/Hr
16	Turn	High	11.49274, 77.88688	15 KM/Hr
17	Blind Spot	Blind Spot	11.49267, 77.88934	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
18	Turn	Medium	11.49284, 77.88915	30 KM/Hr
19	Turn	High	11.49353, 77.86769	15 KM/Hr
20	Turn	Medium	11.49390, 77.86748	30 KM/Hr
21	Turn	Medium	11.49490, 77.86661	30 KM/Hr
22	Turn	Medium	11.52561, 77.85522	30 KM/Hr
23	Turn	Medium	11.52766, 77.85585	30 KM/Hr
24	Turn	Medium	11.53849, 77.85088	30 KM/Hr
25	Turn	Medium	11.55013, 77.85498	30 KM/Hr
26	Turn	High	11.57903, 77.85432	15 KM/Hr
27	Turn	Medium	11.58022, 77.85744	30 KM/Hr
28	Turn	Medium	11.60992, 77.84165	30 KM/Hr
29	Turn	Medium	11.64064, 77.84611	30 KM/Hr
30	Turn	Medium	11.64155, 77.84858	30 KM/Hr
31	Turn	Medium	11.64570, 77.85137	30 KM/Hr
32	Turn	Medium	11.67316, 77.86038	30 KM/Hr
33	Turn	Medium	11.68171, 77.86676	30 KM/Hr
34	Turn	Medium	11.68484, 77.86886	30 KM/Hr
35	Turn	Medium	11.68492, 77.86905	30 KM/Hr
36	Turn	High	11.68472, 77.87281	15 KM/Hr
37	Turn	Medium	11.69280, 77.87439	30 KM/Hr
38	Turn	High	11.69458, 77.87589	15 KM/Hr
39	Turn	High	11.69475, 77.87583	15 KM/Hr
40	Turn	High	11.69494, 77.87544	15 KM/Hr
41	Turn	High	11.69501, 77.87546	15 KM/Hr
42	Turn	High	11.69549, 77.87446	15 KM/Hr
43	Turn	High	11.69915, 77.87478	15 KM/Hr
44	Turn	High	11.69922, 77.87494	15 KM/Hr
45	Turn	Medium	11.76005, 77.89017	30 KM/Hr
46	Turn	Medium	11.76020, 77.89061	30 KM/Hr
47	Turn	Medium	11.75957, 77.89205	30 KM/Hr
48	Turn	High	11.75960, 77.89221	15 KM/Hr
49	Turn	High	11.79280, 77.91433	15 KM/Hr
50	Turn	Medium	11.79281, 77.91501	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
2	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
3	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
4	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
5	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
6	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.58300, 77.85813



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60348, 77.83967



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45543, 77.81740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45563, 77.81708



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49377, 77.88573



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49274, 77.88688



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49267, 77.88934



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49284, 77.88915



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49353, 77.86769



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49390, 77.86748



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49490, 77.86661



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52561, 77.85522



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52766, 77.85585



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53849, 77.85088



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55013, 77.85498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57903, 77.85432



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58022, 77.85744



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60992, 77.84165



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64064, 77.84611



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64155, 77.84858



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64570, 77.85137



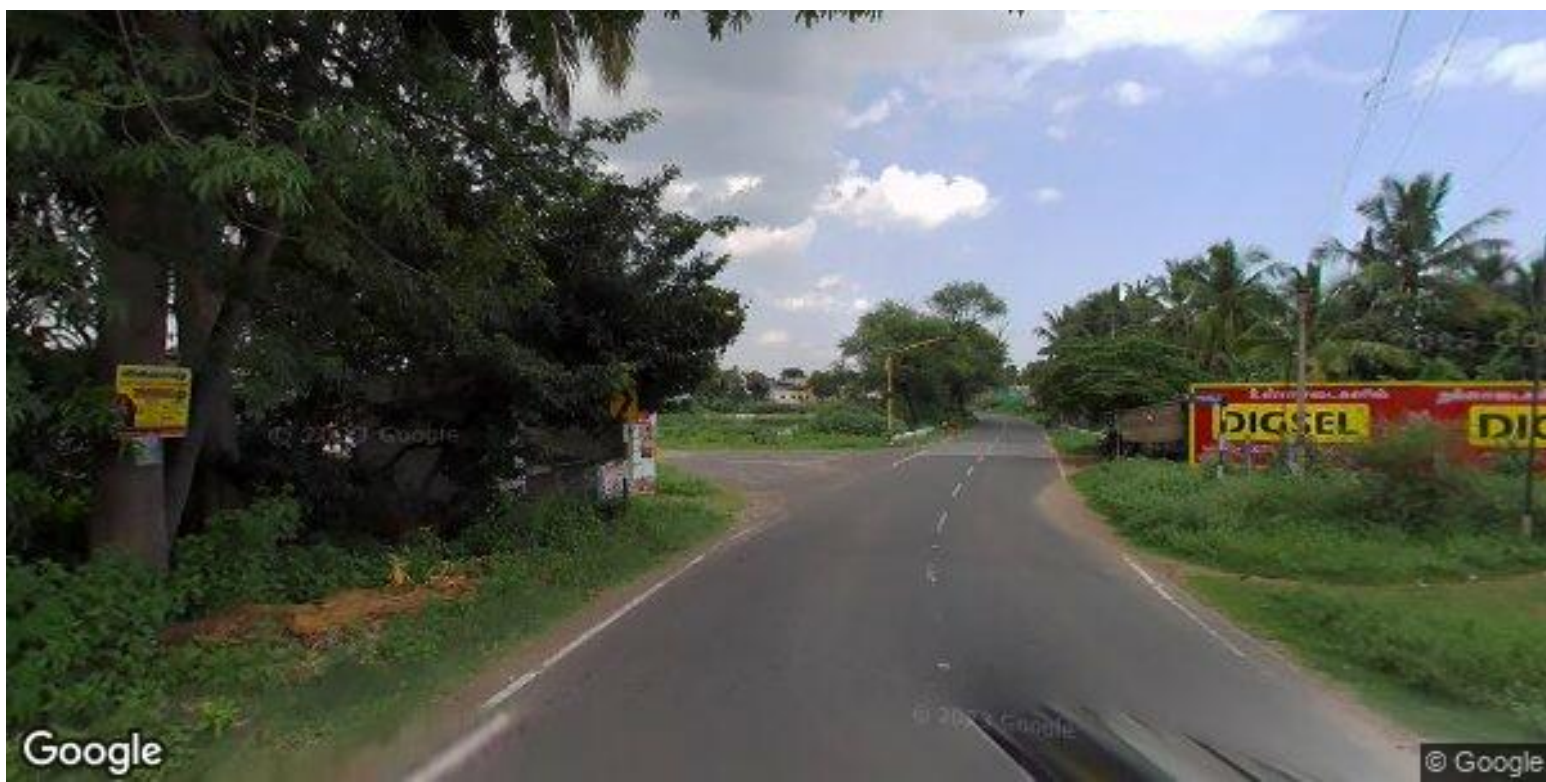
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67316, 77.86038



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68171, 77.86676



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68484, 77.86886



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68492, 77.86905



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68472, 77.87281



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69280, 77.87439



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69458, 77.87589



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

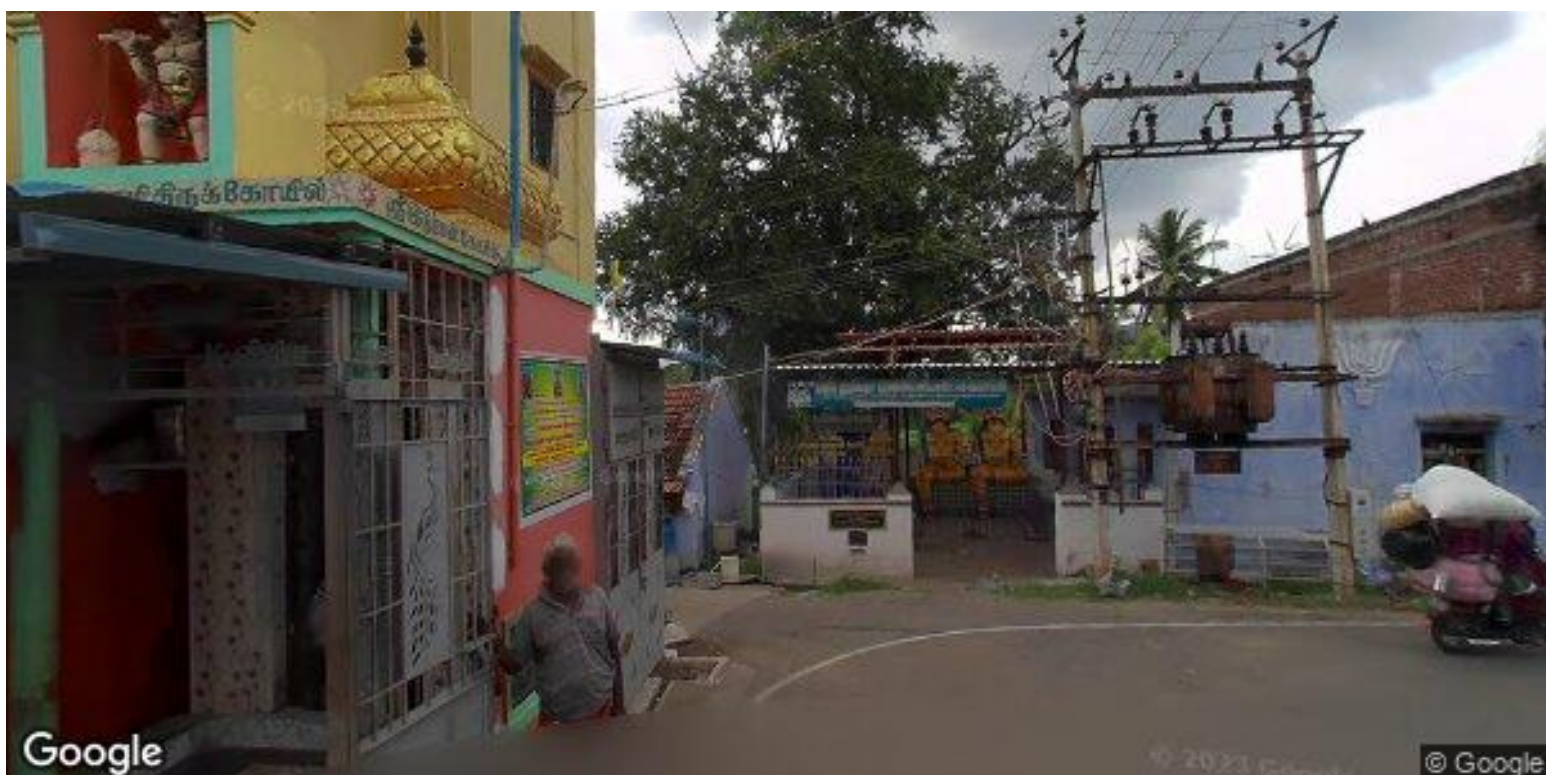
Coordinates: 11.69475, 77.87583



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69494, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69501, 77.87546



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69549, 77.87446



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69915, 77.87478



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69922, 77.87494



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76005, 77.89017



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76020, 77.89061



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.75957, 77.89205



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.75960, 77.89221



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.79280, 77.91433



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.79281, 77.91501
